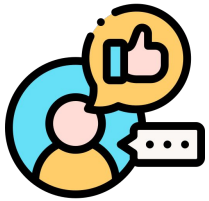
GROUP 7

How Retrieval and Ranking Choices Affect Product Recommendations

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2. Abstract

We study a two-stage product recommender: a retriever selects candidate products, and a ranker orders them. Using the Amazon Reviews 2023 Cell Phones and Accessories category [1], we compare popularity retrieval with a trained two-tower retriever.

We vary candidate count and search settings while keeping the ranker fixed. We also compare popularity order, retrieval score order, an MLP ranker, and one advanced ranker on identical candidate sets.

Finally, we measure candidate recall, final ranking quality, latency, and where missed recommendations arise.

3. Motivation

1. Scale & Architecture

Catalogues are too large for single models

Real catalogues are too large to score with one expensive model. Industrial recommenders use cheap retrieval to narrow down items to hundreds, then a heavier ranker to order them.

2. Candidate Selection

Retrieval creates a hard bottleneck

Retrieval reduces the catalogue to a manageable candidate set. If it misses the relevant product, the ranker cannot recover it.

3. Final Ordering

Position depends on the ranker

Even when the relevant product is retrieved, its position in the final list depends on the ranker.

4. Core Objective

Isolating two-stage failure modes

Studying both stages lets us identify whether a weak recommendation comes from candidate selection or candidate ordering.

4. Task Statement

Task

Build a two-stage retrieve-then-rank product recommender and study how decisions at both stages affect final recommendation quality.

Pipeline

Retrieve (K) eligible products, then order those candidates into a final top-10 list.

Experiment A

Retrieval

Compare retrievers, candidate counts, and search settings while using the same trained ranker.

Experiment B

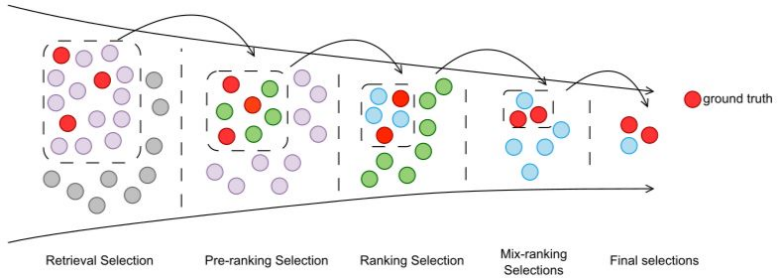
Ranking

Compare ranking methods on identical candidate lists, so differences come from ordering rather than candidate availability.

Output

Evaluation

Trade-off curves of Recall@K, nDCG@10, latency, and a breakdown of retrieval misses versus ranking misses.



5. Proposed Method (Research Questions & Scope)

RQ1 · Retriever design

At a fixed candidate count, does our trained two-tower retriever improve Recall@K over baseline retrieval methods (popularity, BPR-MF, Item-KNN, and SASRec)?

RQ2 · Retrieval and final quality

With the retriever and ranker fixed, how do candidate count K and search settings (Flat, HNSW, IVF) affect Recall@K, final nDCG@10, and latency?

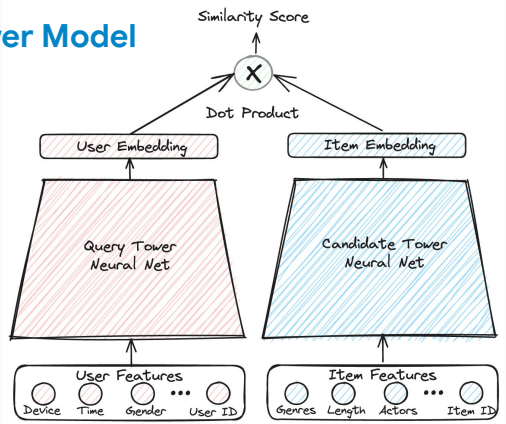
RQ3 · Ranker choice

Given identical candidate lists, how do popularity order, retrieval score order, an MLP, and one advanced ranker (DIN or DCN-V2) compare on nDCG@10?

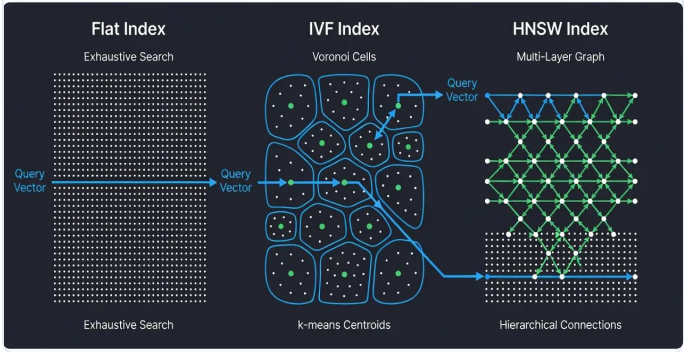
5. Proposed Method (The Retriever)

Component	Description
Baseline (Popularity filtering)	Retrieve products by their positive-review count in the training period.
Item Tower	A frozen sentence encoder embeds each product's title and available subcategory information. A learned projection maps that embedding into the shared retrieval space.
User Tower	A SASRec-style encoder uses the user's earlier product interactions to produce a user vector in the same space.
Training	Train the user tower and item projection on positive interactions using in-batch softmax with logQ sampling correction. [2]
Candidate Generation	Precompute product vectors and index them in FAISS. For each user, retrieve the top (K) products by user–item similarity.

Two-Tower Model

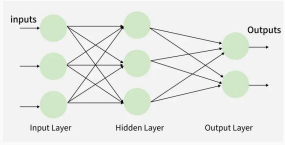


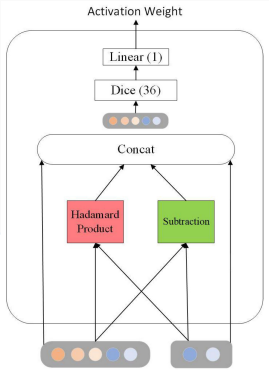
FAISS Vector Indexing

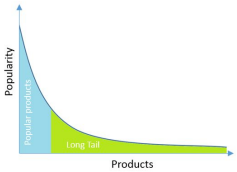


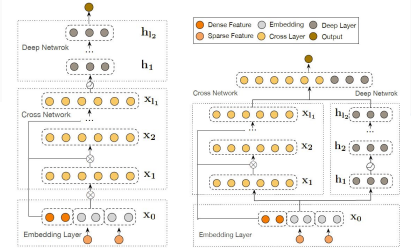
5. Proposed Method (The Ranker)

Component	Description
Input	The (K) products returned by a retriever, together with the user's earlier interactions and available product features.
Primary ranker ⭐	Train an MLP to score each user–product pair. Sort candidates by its score and return the top 10. Train it using candidate lists from a frozen retriever and the 2021 ranker-training period.
Advanced ranker ⭐⭐	Implement **one** additional architecture: DIN-style attention over user history **or** DCN-V2 feature crosses. Select which one to build after the design critique.
Ranking comparison	On identical candidate lists, compare: - popularity order - retrieval score order - the MLP - the selected advanced ranker using nDCG@10 and ranking latency.









6. Progress (1 of 2)

Dataset Profile (Cell Phones and Accessories Category)

Reviews: **2,752,785** • Users: **380,999** • Products: **111,480**

Filtering Rules

- **Deduplication:** Retained only the earliest review for repeated user–product reviews.
- **Release-time exclusion:** Removed products whose first review in the dataset comes after the target's review from the candidate ranking.

Period	Use	Reviews
Up to 26 Jul 2020	Retriever training	**1,877,548**
26 Jul 2020 - 11 Aug 2021	Ranker training	**331,332**
11 Aug 2021 - 16 Jul 2022	Validation	**239,437**
From 16 Jul 2022	Test	**304,468**

6. Progress (2 of 2)

Completed: Benchmarked 4 baselines, evaluated on the data from the test set.

Provides an initial reference showing the performance of the four baseline models.

Model	recall@10	ndcg@10	recall@100
BPR	0.0030 ± 0.0002	0.0015 ± 0.0001	0.0172 ± 0.0004
ItemKNN	0.0014	0.0008	0.0052
Popularity	0.0022	0.0010	0.0166
SASRec	0.0053 ± 0.0006	0.0026 ± 0.0003	0.0248 ± 0.0011

7. Proposed Evaluation

Experiment Design

We will be using a chronological split. We hold the ranker fixed when comparing retrievers and use identical candidates when comparing rankers. Features and item statistics use only data available before the test event.

Stage	Metrics
Retrieval	Recall@{10, 50, 100, 200, 500}
Ranker	nDCG@10 on cases where the target was retrieved; use identical candidate lists for every ranker
End to end	Recall@10, nDCG@10; retrieval-miss and ranking-miss counts
Beyond accuracy	Catalogue coverage, popularity concentration of the top-10 (if time permits)
System	Retrieval, ranking, and total latency per request

8. Resources

Data / Corpora

Dataset

Amazon Review 2023 dataset (Cell Phones & Accessories)

Interaction Signals

Reviews & ratings

Features

Product metadata (title, category, price)

Compute

Environment

NUS SoC Computer Cluster

Execution

Command-line scripts

Software

Existing Libraries

RecBole (baselines), **sentence-transformers** (embeddings), **FAISS** (search)

Custom PyTorch Implementation

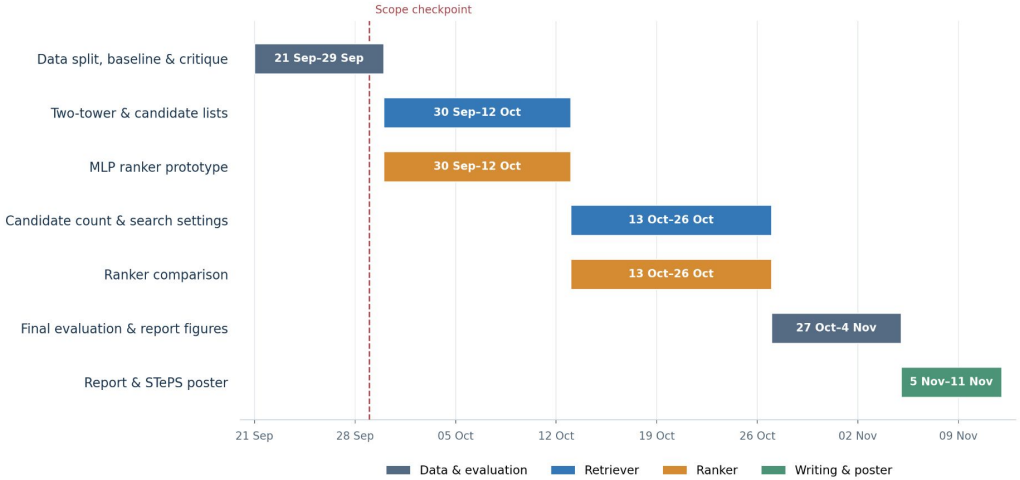
Retriever, ranker, and cascade experiments

Replicability

Split-generation script, per-experiment config files, fixed seeds, & shared evaluation script

9. Schedule & Role Assignment

Schedule & Milestones



Role Assignments

Phua Qing Rong
Data preparation, shared evaluation, and failure analysis

Brandon Kang
Two-tower retriever, exact search, and FAISS experiments

Tong Jing Yen
Ranker training and comparisons on fixed candidate lists

All members
Interpret results, write the report, and prepare the poster

10. Acknowledgements

We acknowledge the creators of the Amazon Reviews 2023 dataset and the maintainers of the open-source software used in this project. We will add individual acknowledgements for advice or assistance actually received before the final submission.

11. References

[1] arXiv 2024

‘Bridging language and items for retrieval and recommendation’

Hou, Y. et al. (2024) — *arXiv preprint*, arXiv:2403.03952

https://doi.org/10.48550/arXiv.2403.03952

[2] RecSys 2019

'Sampling-bias-corrected neural modeling for large corpus item recommendations'

Yi, X. et al. (2019) — *Proceedings of RecSys '19*, pp. 269–277.
https://research.google/pubs/sampling-bias-corrected-neural-modeling-for-large-corpus-item-recommendations/

Do not include this slide as part of your submission. This for your group's information only.

Overview

The Project Critique is a key milestone in your project, where you will get feedback from the teaching staff on your project. The update should briefly describe your team's project scope, along with a description of your proposed method and progress towards your goal. You have made along with any preliminary results you may have.

You may clone this document as a template but this is indicative only; feel free to embellish. Keep your slides in this widescreen format (16:9 aspect ratio)

Submitting your Project Critique Slides

Your update needs to be submitted in Canvas:

1. Canvas Assignments – only one student in the group should submit the slide deck in .pdf form.

The same student who submitted the original slides should submit any revised version. You may submit revisions as many times as you want before the PDF submission deadline on Mon, 29 Sep 2026, 23:59 SGT.

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Grading

Your group's project critique will be marked by one of the instructors and by your assigned project mentor (a teaching assistant).

Feedback from both parties will be transmitted to each group member through Canvas Grades. We aim to provide feedback within two weeks.

This is a **formative assessment**, so please take the feedback as constructive criticism to try to address the weaknesses identified.

The Project Critique constitutes **5%** of your overall marks for your course grade. The comments should help guide the development of your project and serve as a talking point with your instructing staff.

Do not include this slide as part of your submission. This for your group's information only.

Outline

1. Title
2. Abstract
3. Motivation
4. Task Statement
5. Proposed Method
6. Progress
7. Proposed Evaluation
8. Resources
9. Schedule and Role Assignment

10. Acknowledgements
11. References

You need not show an outline slide in your group's **Project Critique**; this is given here just to help to show what sections are mandatory for the update.

Maximum 20 slides
(exclusive of Acknowledgements and References; unlimited slides for these sections)

Do not include this slide as part of your submission. This for your group's information only.

Marking Rubric

1. Presentation (30%)
   - Motivation
   - Structure & Layout
   - Appropriate images/plots/figures

2. Methodology (60%)
   - Good understanding of task & dataset
   - Resources & related works (if applicable)
   - Overview to proposed approach
     (e.g., methods, techniques, frameworks, tools, etc.)
   - Current results (if applicable)
   - Current challenges and open questions

Marking for the Project Critique will follow this template.

3. Administration (10%)
   - Distribution of workload
   - Next steps (basic timeline & milestones)

Version History

V1.0 260915 – Initial version, adopted from CS4248.